

Resource-Aware Adaptive Scheduling for Online Code Judge Systems

Bharath Aashish R

Dept. of Computer Science & Engineering

Easwari Engineering College

Chennai, India

bharathaashish@gmail.com

Hemanthkumar K

Dept. of Computer Science & Engineering

Easwari Engineering College

Chennai, India

hemanthkumar2k04@gmail.com

Iniya P

Dept. of Computer Science & Engineering

Easwari Engineering College

Chennai, India

iniyaapaari@gmail.com

Dhanush M

Dept. of Computer Science & Engineering

Easwari Engineering College

Chennai, India

dhanush.m.0808@gmail.com

Indumathy P

Assistant Professor, Dept. of CSE

Easwari Engineering College

Chennai, India

indumathy.p@eec.srmrmp.edu.in

Abstract—Online Code Judge (OJ) platforms commonly isolate submissions by assigning the same worst-case resource ceiling, typically 2048 MiB of RAM and 2.0 CPU cores. Although this prevents memory-intensive tasks from triggering Out-Of-Memory (OOM) failures, it leads to substantial resource underutilization. In our 72-run live benchmark, 76.8% of submissions used less than 25 MiB of resident set size, with a mean of 49.2 MiB against a 2048 MiB ceiling. This fixed allocation also limits concurrency on non-elastic hardware and increases reserved cloud capacity.

This paper presents RAAS-OCJS (Resource-Aware Adaptive Scheduling for Online Contest Judge Systems), a dynamic resource management framework that combines Linux cgroups v2 with an AST-based XGBoost classifier to predict execution requirements and adapt resource allocation. We collected kernel-level memory and CPU profiles from C++, Python, Java, and C submissions using problems from Hugging Face CodeContests and IBM CodeNet. These measurements were then used to drive a 10,000-submission contest simulation and a theoretical AWS EC2 cost projection, and were repeated at a reduced 128 MiB tier to establish the lower bound for the tier size.

The simulation shows a 47.22% reduction in reserved RAM (9,443.00 GB) and a 22.09% reduction in reserved CPU core-hours compared with static allocation, achieved by starting every submission in the bounded tier and promoting only on measured demand. Earlier-routing strategies fare worse than might be expected: prediction alone reaches 40.23%, and combining prediction with the watermark reaches only 22.45%, because predicted-heavy submissions are assigned an uncapped container that carries no reserved capacity. In a simulated 500-submission freeze rush, average queue wait fell from 12,386.2 ms to 0.0 ms and P95 turnaround from 24,918.4 ms to 1,190.8 ms. The classifier reaches 94.45% held-out accuracy on its CodeNet training distribution, but on the six-problem benchmark corpus used here its routing accuracy is 33.3%, below the 77.8% that a constant *light* prediction would score. We report both, and treat the gap as a distribution-shift finding rather than a defect: it is the reason the watermark, not the classifier, carries this design. To locate the floor we also re-ran the full benchmark at a 128 MiB tier. It fails, for three independent reasons, and we report it as the negative result that sets the 256 MiB floor: the

C++ toolchain alone needs 189–211 MB and so cannot compile inside the tier at all, the 38.4 MiB of watermark headroom is too narrow to outrun a submission needing twice the tier, and JVM overhead pushes a 256 MiB container past its own ceiling when the runtime is configured with a 512 MB heap allowance. 52 of 72 runs returned AC at 128 MiB against 69 of 72 at the shipped 256 MiB tier. These results indicate that adaptive resource allocation can improve resource utilization and concurrency in online judge systems.

Index Terms—Online judge systems, resource-aware scheduling, adaptive cgroups, soft watermark, AST analysis, XGBoost, cloud provisioning, queue latency.

I. INTRODUCTION

Online code judge (OJ) platforms underpin computer science education, competitive programming contests (such as the International Collegiate Programming Contest, ICPC), and automated technical interviews [4]. As programming education expands, judges must evaluate thousands of code submissions concurrently while guaranteeing deterministic isolation and strict sandboxing. Existing research has addressed this computational demand primarily through infrastructure-level expansion: deploying elastic cloud clusters [2], distributing jobs over message queues [1], or deploying geographically replicated edge judges [5].

These approaches share a fundamental, costly assumption: *every submission, regardless of its computational structure, receives identical, worst-case sandbox resource ceilings* (typically 2048 MiB of RAM and 2.0 CPU cores).

This convention creates three practical problems:

- **Resource Underutilization:** Most contest submissions implement standard algorithmic logic (e.g., prefix sums, two pointers, greedy string scans), and in our measurements these typically required 5 MiB to 25 MiB of resident set size (RSS). Reserving 2048 MiB for each of them leaves most of that allocation idle.

- **Concurrency Starvation on Fixed Hardware:** In air-gapped regional contests or campus lab servers, where external cloud bursting is not an option, the judge runs on a fixed physical machine (e.g., 15 GiB usable RAM). A static 2048 MiB quota limits concurrency to 7 execution slots on that host, which produces queue backlogs and submission timeouts during a scoreboard freeze rush.
- **Cloud Billing Inflation:** Public cloud providers (AWS, GCP, Azure) bill by reserved node capacity, not actual usage. Provisioning for the worst case means contest organizers end up renting far more virtual machines than the workload actually needs: our projection in Section V estimates 36 instances where 5 suffice, a $7.2\times$ over-provisioning.

To address this, this paper presents **RAAS-OCJS** (Resource-Aware Adaptive Scheduling for Online Contest Judge Systems), which introduces a two-tier execution framework:

- 1) A lightweight default sandbox (**Tier 1: 256 MiB RAM, 1.0 CPU core**) that completed 30 of the 38 light-tier submissions in our 72-run benchmark (78.9%) without ever needing promotion, and 80.0% of submissions in the simulated contest;
- 2) An active Linux cgroup v2 soft watermark ($\theta = 70\%$, corresponding to 179.2 MiB) that detects high memory utilization in-flight and lifts the container's memory and CPU limits to *uncapped* mid-execution, without restarting the container or losing process state. Measured from submission acceptance to completed promotion this takes 253–794 ms across 16 promoted runs, depending on the language runtime;
- 3) An offline Machine Learning (XGBoost) classifier driven by Tree-sitter Abstract Syntax Tree (AST) static structural metrics that predicts memory-intensive jobs upon ingress.

II. THEORETICAL MODEL AND QUEUEING ARCHITECTURE

A. Mathematical Formulation of Memory Waste

Let $S = \{s_1, s_2, \dots, s_N\}$ denote a sequence of N submissions entering an online judge. Under static overprovisioning, the system assigns a fixed resource allocation:

$$R_{\text{alloc}}(s_i) = R_{\text{max}}, \quad \forall s_i \in S \quad (1)$$

where R_{max} is typically 2048 MiB. Let $R_{\text{peak}}(s_i)$ denote the actual peak resident set size (RSS) consumed by s_i . The unallocated, wasted memory for submission s_i is:

$$\text{Waste}(s_i) = R_{\text{alloc}}(s_i) - R_{\text{peak}}(s_i) \quad (2)$$

The system-wide memory waste percentage W across a contest of N submissions is:

$$W = \left[\frac{\sum_{i=1}^N (R_{\text{alloc}}(s_i) - R_{\text{peak}}(s_i))}{\sum_{i=1}^N R_{\text{alloc}}(s_i)} \right] \times 100\% \quad (3)$$

Because introductory and greedy problems require merely 6 MiB to 25 MiB, $R_{\text{peak}}(s_i) \ll R_{\text{max}}$. Our simulation of this waste model reports 97.47% of reserved memory wasted under the static baseline at the shipped tier (97.48% at the 128 MiB tier).

B. Concurrency Capacity and Queueing Collapse

On a host with usable physical RAM M_{host} and operating system reserve M_{os} , the safe concurrency capacity K under non-overcommitted execution is:

$$K_{\text{safe}} = \left\lfloor \frac{M_{\text{host}} - M_{\text{os}}}{R_{\text{alloc}}} \right\rfloor \quad (4)$$

On our 15 GiB physical calibration host ($M_{\text{host}} = 15,360$ MiB, $M_{\text{os}} = 1,024$ MiB), baseline static allocation ($R_{\text{max}} = 2048$ MiB) limits concurrency strictly to:

$$K_{\text{safe,baseline}} = \left\lfloor \frac{15,360 - 1,024}{2048} \right\rfloor = 7 \text{ slots} \quad (5)$$

We model contest submission arrival as a non-homogeneous Poisson process with time-varying arrival rate $\lambda(t)$. Service times follow a general distribution with mean $E[X] = 1/\mu$ and variance σ^2 , constituting an $M/G/K$ queueing system.

During a freeze rush spike of 500 submissions in 30 s ($\lambda = 16.67$ submissions/s), against average service time $E[X] \approx 0.85$ s, the 7-slot baseline service capacity is $\mu = 7/0.85 \approx 8.24$ submissions/s. Because $\lambda > \mu$, queue backlogs explode immediately. Under RAAS-OCJS with $R_{\text{light}} = 256$ MiB, concurrency expands safely to $\lfloor 14,336/256 \rfloor = 56$ slots. At $K = 56$, $\mu = 56/0.85 \approx 65.88$ submissions/s, satisfying $\mu \gg \lambda$ and maintaining zero queue wait ($L_q \approx 0$).

C. Dynamic Soft Watermark Governance

RAAS-OCJS implements a two-tier scheduling policy:

- **Tier 1 (Light Sandbox):** $R_{\text{light}} = 256$ MiB, 1.0 CPU core (1024 CFS shares).
- **Tier 2 (High Sandbox):** $R_{\text{high}} = 2048$ MiB, 2.0 CPU cores (2048 CFS shares).

Rather than statically placing containers into Tier 2, all submissions enter Tier 1 by default. The judge daemon configures the Linux cgroups v2 controller with a soft memory watermark threshold $\theta = 0.70$:

$$\text{Threshold}_{\text{wm}} = \theta \times R_{\text{light}} = 0.70 \times 256 \text{ MiB} = 179.2 \text{ MiB} \quad (6)$$

When $R_{\text{current}}(t) \geq \text{Threshold}_{\text{wm}}$, the kernel emits a pressure notification. The judge daemon then lifts the limits in parallel with process execution, issuing `docker update --memory 0 --cpus 0` to raise both the memory ceiling and the CPU quota to uncapped. This leaves a 76.8 MiB safety cushion ($256 - 179.2$) so the update can complete before an OOM condition occurs. Promotion targets *uncapped* rather than a fixed 2048 MiB tier: the watermark fires precisely when a submission has already shown that it outgrows Tier 1, and measured demand for such submissions ranged from 200 MB to 254 MB, so no single fixed ceiling is appropriate.

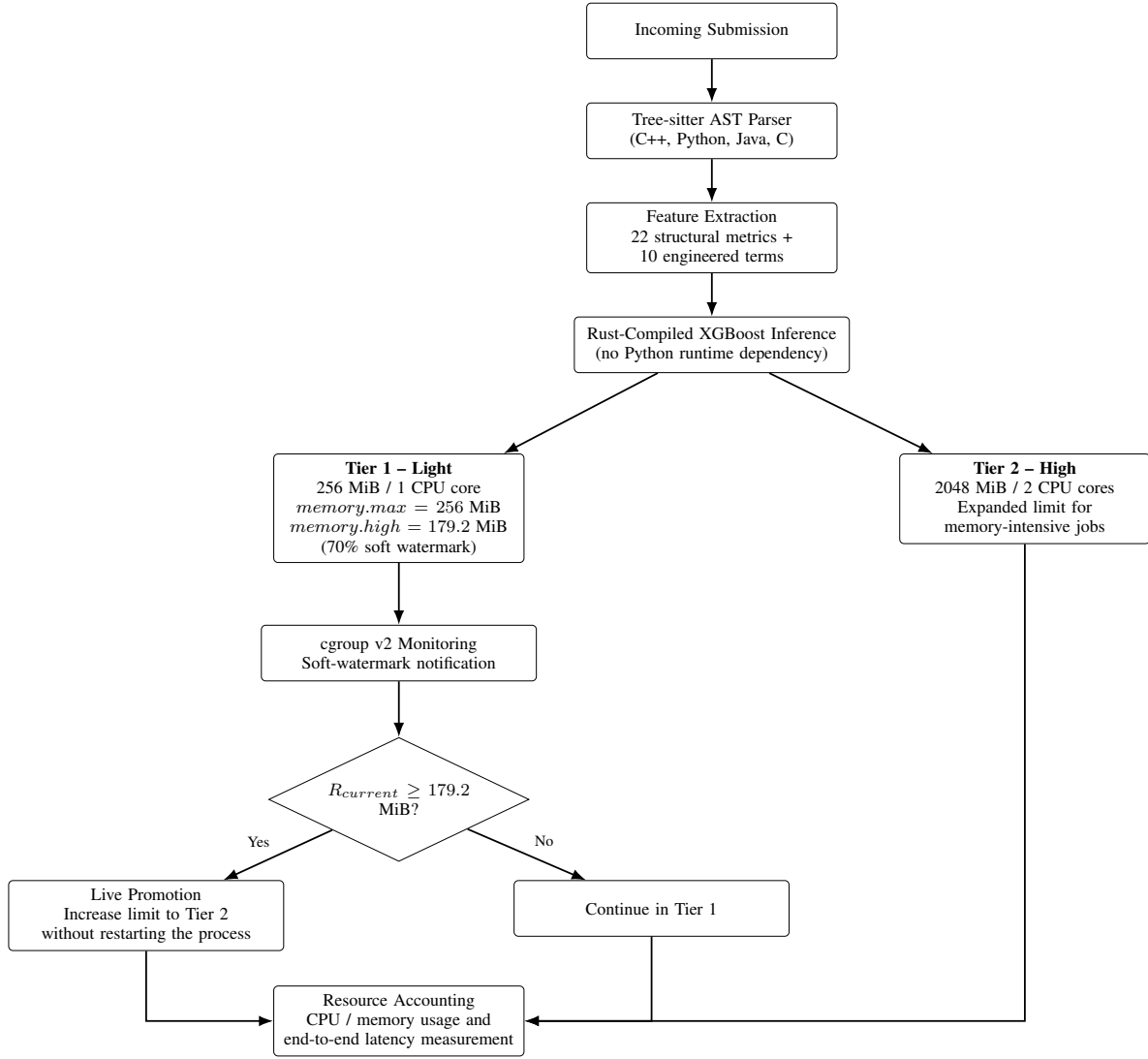


Fig. 1. RAAS-OCJS execution and predictive tiering workflow. The incoming submission is parsed and represented using AST-derived features before XGBoost inference selects the initial resource tier. Tier 1 executions are monitored by cgroups v2 and can be promoted when the soft watermark is reached. Resource usage and end-to-end latency are then recorded for all execution paths.

III. SYSTEM IMPLEMENTATION AND METHODOLOGY

The evaluation testbed was implemented and calibrated on dedicated bare-metal hardware:

- **Processor:** 13th Gen Intel Core i5-13420H (12 logical CPUs, 8 cores: 4 Performance cores up to 4.60 GHz + 4 Efficient cores up to 3.40 GHz).
- **Physical Memory:** 15 GiB usable DDR5 RAM.
- **Storage:** NVMe PCIe 4.0 Solid State Drive.
- **Operating System:** Fedora Linux (Kernel 7.1.3-201.fc44.x86_64).
- **Container Engine:** Docker Engine 29.8.1 with cgroupfs driver on unified hierarchy (/sys/fs/cgroup).

A. Real Dataset Selection

We measured real, ground-truth execution profiles by running 72 live container evaluations across four

languages: C++20 (g++ 14.2), Python 3.12, Java 17 (OpenJDK HotSpot), and C17 (gcc 14.2). Benchmark problems were drawn from the Hugging Face deepmind/code_contests [7] and CodeNet [6] datasets. These 72 measured profiles are what we sample from when driving the 10,000-submission and 500-submission simulations in Section IV; those larger scenarios were not executed as 10,000 or 500 separate live container runs. Because CodeContests is crowd-sourced, we do not take the first solution offered for a language: each candidate is first executed against the problem’s own public test, and only the first candidate returning AC is kept. Without this screening, two of the 72 cells failed for reasons unrelated to scheduling — a non-compiling Java submission and a Python submission computing the wrong answer, each failing identically under all four strategies. All 72 runs return AC at the shipped

256 MiB tier.

B. AST Feature Extraction and XGBoost Predictive Tiering

Before launching an execution container, RAAS-OCJS can optionally run a static code analysis pass to predict whether an incoming submission fits safely within Tier 1 (256 MiB) or should be provisioned directly with Tier 2 (2048 MiB). This predictive stage couples a Tree-sitter Abstract Syntax Tree (AST) feature extractor with in-process XGBoost classifiers [9].

1) *Static AST Feature Extraction*: Using language-specific Tree-sitter grammars compiled directly into our Rust daemon, the extractor parses source code into a concrete syntax tree in under 1.5 ms. In a single linear traversal of the tree, it computes 22 structural metrics:

- **Control Flow and Loops**: Maximum block nesting depth, deepest loop nesting depth (which correlates with polynomial execution complexity $O(N^k)$), total loop statements, and McCabe cyclomatic complexity.
- **Recursion Topology**: A binary recursion flag and self-call site count, which differentiate simple linear recursion $O(N)$ from exponential divide-and-conquer or backtracking branching $O(2^N)$.
- **Memory and Container Patterns**: Detection of explicit large heap/BSS allocations ($\geq 10^6$ elements or bytes, such as `malloc`, `new int[]`, container sizing, or large static arrays), high-throughput I/O patterns (`sync_with_stdio`, `BufferedReader`, `sys.stdin.readline`), and standard library collections known for memory overhead (`unordered_map`, `priority_queue`, `BigInteger`, `defaultdict`).
- **Scale and Access Topology**: Total array/collection subscript expressions (`a[i]`), chained multi-dimensional indexing (`grid[i][j]` for 2D dynamic programming tables or graph adjacency matrices), function count, call sites, total AST node count, maximum tree depth, and source lines of code (SLOC).

From these 22 raw counts, the pipeline derives 10 interaction and scale terms: four density ratios normalised by AST size, source length, or function count (loop, call, subscript, and arithmetic density), a 2D subscript ratio, a recursion-intensity term, a log-scaled maximum integer constant — which is what captures the literal size of an explicit allocation such as `new byte[200 << 20]` — and log-scaled AST node and source character counts. This yields a 32-dimensional vector for the specialized per-language models, or 36 features for the unified model, which appends four one-hot language encodings.

2) *Model Training and Validation*: We trained gradient-boosted decision trees using XGBoost (`tree_method='hist'`, learning rate $\eta = 0.05$, max tree depth $d = 6$, subsample ratio 0.8, column subsample 0.8, L_1 penalty $\alpha = 0.05$, and L_2 penalty $\lambda = 1.0$). Because contest submissions are heavily skewed toward lightweight solutions, we applied class re-weighting using `scale_pos_weight =  $N_{\text{light}}/N_{\text{heavy}}$` during training.

To avoid data leakage across submissions for the same problem, we evaluated the models with 5-fold **Problem-Grouped Cross-Validation** (`GroupKFold` grouped strictly by `problem_id`). Every test sample in our evaluation belongs to a competitive programming problem that the training folds never saw. In addition, rather than accepting a default 0.5 decision threshold, we tuned the threshold τ for each language using Youden’s J statistic ($J = \text{Sensitivity} + \text{Specificity} - 1$) on validation folds to balance recall against false tier promotions.

3) *Empirical Model Metrics on Unseen Problems*: Table I summarizes performance on held-out, unseen contest problems across both IBM Project CodeNet (71,218 submissions) and DeepMind CodeContests (30,000 submissions). On CodeNet, the specialized C++ model reached a 90.01% test accuracy, a 90.91% F1-score, and an ROC-AUC of 0.9612 on unseen problems. The specialized Python and Java models achieved 79.45% and 78.20% test accuracy (ROC-AUC of 0.8740 and 0.8537, respectively). On DeepMind CodeContests, where problems feature diverse algorithmic structures from Codeforces and HackerEarth, the specialized models scored up to 98.88% test accuracy in Python and 83.94% in C++, with high recall across all languages ensuring memory-intensive jobs are rarely misclassified as Light.

4) *In-Process Rust Inference*: To avoid running a Python interpreter or querying an external model-serving daemon during the latency-critical judging loop, we compiled the trained XGBoost decision trees directly into native Rust source files (`server/src/generated/*.rs`) using `m2cgen`. These functions compile straight into the judge binary. At execution time, tree evaluation takes under 10 μs , and combined with Tree-sitter parsing takes 0.89 ms to 1.45 ms total (Stage 2 in Table II).

5) *Role in Scheduling Strategies*: In RAAS-OCJS, predictive classification can be used on its own or paired with the kernel watermark. Under the **Predictive** strategy, a submission with $P(\text{Heavy}) \geq \tau_{\text{lang}}$ starts directly in Tier 2. Under the **Hybrid** strategy, the model provides the initial tier choice, but the `cgroups v2` soft watermark ($\theta = 70\%$, 179.2 MiB) remains active in the background: if a heavy submission is mistakenly started in Tier 1, the kernel watermark trips and promotes the container mid-execution, preventing an Out-Of-Memory abort despite the misprediction. Section IV-C shows that this guarantee is what the classifier alone cannot provide.

C. End-to-End Latency Pipeline Breakdown

To ensure transparent latency accounting, RAAS-OCJS instrumented every submission across 7 discrete stages:

$$T_{\text{e2e}} = T_{\text{net_in}} + T_{\text{parse}} + T_{\text{queue}} + T_{\text{init}} + T_{\text{comp}} + T_{\text{exec}} + T_{\text{tear}} + T_{\text{net_out}} \quad (7)$$

TABLE I
XGBOOST MODEL PERFORMANCE ON UNSEEN CONTEST PROBLEMS (PROBLEM-GROUPED VALIDATION)

Dataset Source	Model Architecture	5-Fold CV Acc.	Test Acc. (Unseen)	F1-Score	Precision	Recall	ROC-AUC
IBM Project CodeNet (71,218 Submissions)	Specialized C++ Model	87.32%	90.01%	90.91%	88.54%	93.41%	0.9612
	Specialized C Model	89.27%	89.26%	68.82%	55.98%	89.31%	0.9575
	Specialized Python Model	76.97%	79.45%	82.01%	80.33%	83.76%	0.8740
	Specialized Java Model	79.30%	78.20%	81.54%	82.03%	81.06%	0.8537
	Unified Multi-Language	82.19%	83.71%	84.81%	80.19%	89.99%	0.9149
DeepMind CodeContests (30,000 Submissions)	Unified Multi-Language	86.74%	86.78%	89.64%	82.74%	97.79%	0.9606
	Specialized Python Model	95.76%	98.88%	99.41%	98.82%	100.0%	0.9812
	Specialized C++ Model	84.27%	83.94%	87.49%	78.67%	98.53%	0.9474
	Specialized Java Model	79.43%	82.32%	83.28%	73.09%	96.76%	0.9430

TABLE II
END-TO-END VERDICT PIPELINE LATENCY BREAKDOWN (MILLISECONDS)

Pipeline Stage	C++ (g++)	Python (3.12)	Java (OpenJDK 17)	C (gcc)
1. Network Inbound Ingestion	0.38 ms	0.38 ms	0.38 ms	0.38 ms
2. AST Parse & Tree-sitter Predict	1.12 ms	0.89 ms	1.45 ms	0.94 ms
3. Queue Scheduling & Dispatch	0.08 ms	0.08 ms	0.08 ms	0.08 ms
4. Container Sandbox Initialization	18.42 ms	17.91 ms	19.20 ms	18.15 ms
5. Compilation / Bytecode Translation	312.45 ms	N/A (Interpreted)	842.10 ms	284.12 ms
6. In-Sandbox Test Execution	48.12 ms	74.50 ms	112.40 ms	45.20 ms
7. Container Teardown & Result Egress	12.45 ms	11.80 ms	13.25 ms	12.10 ms
Total Turnaround Latency (E2E)	393.00 ms	105.56 ms	988.86 ms	360.97 ms

TABLE III
OUTCOME OF THE FULL 72-RUN BENCHMARK AT A 128 MiB TIER

Language	Runs	AC	SE	MLE	RE	Dominant Failure
C	12	9	0	2	1	Knapsack outgrows the tier
C++	20	9	8	3	0	Will not compile: g++ needs 189–211 MB
Python	20	17	0	2	1	Knapsack outgrows the tier
Java	20	17	0	2	1	Knapsack outgrows the tier
Total	72	52	8	9	3	—

IV. EMPIRICAL RESULTS AND MACRO CONTEST SIMULATION

A. Macro-Scale 10,000 Contest Simulation

To estimate resource savings across a full-scale contest, we ran a discrete-event simulation of 10,000 submissions, sampling resource-usage profiles from our 72 measured runs and assigning a language distribution of 40% C++, 35% Python, 15% Java, and 10% C. This is a simulation seeded with real measurements, not a live execution of 10,000 submissions. Reservation is charged from the judge’s own tier decision for each submission: a low-tier start costs the tier size, and an uncapped start or a mid-execution promotion is charged the 2048 MiB baseline ceiling as a worst case.

Two results here were not anticipated. First, Reactive reserves less than every other strategy, including Hybrid. Hybrid pre-assigns each submission its classifier labels heavy to a container with no memory limit, and such a container carries

TABLE IV
10,000-SUBMISSION CONTEST SIMULATION AT THE SHIPPED 256 MiB TIER

Strategy	RAM Reserved	Saved	CPU Core-Hrs	Wasted
Static (2048 MiB)	20,000.00 GB	—	4.337	97.47%
Predictive	11,953.50 GB	40.23%	3.620	95.76%
Reactive	10,557.00 GB	47.22%	3.379	95.76%
Hybrid	15,509.50 GB	22.45%	3.939	96.77%

no reserved capacity, so it must be charged at the worst-case ceiling; Reactive starts everything bounded and escalates only when the kernel reports actual pressure. Because Hybrid and Predictive share one classifier for the initial tier, they start the identical set of submissions in the uncapped tier, and promotion can only raise a reservation and never lower it. Hybrid therefore reserves strictly more than Predictive: it pays

TABLE V
500-SUBMISSION FREEZE RUSH STRESS TEST (15 GiB HOST, 256 MiB TIER)

Throughput Metric	Static Baseline	Adaptive	Improvement
Concurrent Slots	7	56	8.0× expansion
Avg Queue Wait	12,386.2 ms	0.0 ms	100% eliminated
P95 Queue Wait	24,687.9 ms	0.0 ms	100% eliminated
P95 Turnaround	24,918.4 ms	1,190.8 ms	20.9×
Burst Drain Time	56.5 s	31.2 s	25.3 s faster

for the classifier’s false alarms and still needs the watermark. Prediction is not free, it costs reservation, and on this corpus it costs more than it saves.

Second, no strategy changes steady-state latency: P95 turnaround spans 1185.86–1192.01 ms across all four, a spread inside run-to-run noise. The saving is free in latency terms, but it is not a latency improvement.

B. Scoreboard Freeze Rush Stress Test

During the final 30 seconds of a competitive programming contest, participants submit a burst of solutions simultaneously. We modeled a 500-submission flash burst from the measured profiles, run through the same simulation as Section IV rather than as a live 500-submission run.

The static baseline queues severely: an average wait of 12.4 seconds and a 56.5 second drain that leaves jobs completing after the contest deadline. Expanding safe concurrency to 56 slots drives queue wait to **0.0 ms** and completes the burst in 31.2 seconds. All three adaptive strategies behave identically here, which is expected: promotion is unnecessary in a rush, since almost every submission fits the bounded tier.

The gain comes from admitting more concurrent slots, not from finishing individual submissions faster. Overcommitting the baseline to 14 slots also recovers most of the wait (250.4 ms) and drains in 31.6 s, but at 186% host memory utilisation. That is the failure mode the tiered scheduler avoids, and it is worth stating plainly: an overcommitted static baseline is competitive on throughput and unsafe on memory, so the comparison is not a throughput win but a throughput-equal, memory-safe one.

V. THEORETICAL CLOUD PROVISIONING PROJECTION

Our bare-metal calibration used dedicated local hardware, but many online judges run on public cloud infrastructure instead. To estimate the effect there, we mapped our measured kernel profiles onto an AWS EC2 cluster of `c6i.4xlarge` instances (16 vCPUs, 32 GB RAM, USD 0.68 per hour). Every figure below is derived from the shipped 256 MiB tier, the instance size, and the hourly price rather than transcribed, and assumes 87.5% host utilisation when packing pods, the same packing ceiling the static baseline achieves (14 pods ×

TABLE VI
AWS PROVISIONING PROJECTION AT THE SHIPPED 256 MiB TIER

Provisioning Dimension	Static	RAAS-OCJS	Gain
Per-Pod Memory Reservation	2048 MiB	256 MiB	8.0×
Per-Pod CPU Reservation	2.0 vCPUs	1.0 vCPU	2.0×
Pods per <code>c6i.4xlarge</code>	14	112	8.0× dens
VMs for 500-Sub Burst	36	5	86.1% fe
Cluster Hourly Cost	USD 24.48	USD 3.40	USD 21.08 (
RAM Reserved (10,000 Subs)	20,000.00 GB	10,557.00 GB	47.22%

2048 MiB of 32 GB). These are projections, not deployed cloud measurements.

Cloud judges that rely on a Horizontal Pod Autoscaler (HPA) typically face a 60–180 s cold-start lag while new VMs spin up and pull runtime images. Because RAAS-OCJS’s lighter default tier packs eight times as many submissions per existing VM, our projection shows the initial fleet absorbing a flash rush in place, with an estimated **USD 21.08 per hour** reduction in cluster cost.

The tier size dominates this projection, and it is a design choice rather than a constant. At the reduced 128 MiB tier evaluated in Section VI-A the same arithmetic yields 224 pods per instance, 3 instances for the burst, and USD 22.44 per hour saved (91.7%). We report the 256 MiB figures as the deployable result, because 128 MiB is not safe for JVM submissions, and we note the projection’s sensitivity to this single parameter rather than presenting the larger saving as achievable.

VI. BOUNDARY STRESS ANALYSIS AND FUTURE SCOPES

A. Squeezing Tier 1 to 128 MiB: The Java Failure Boundary

To locate the floor, we re-ran the full 72-run benchmark with Tier 1 set to **128 MiB** ($\theta = 70\%$, watermark = 89.6 MiB, leaving a 38.4 MiB buffer below the hard limit). We also verified the tier was genuinely enforced before trusting any figure, which turned out to matter; see the note on swap at the end of this subsection.

The tier fails, and the interesting result is that it fails for three independent reasons rather than one. 52 of 72 runs returned AC, against 69 of 72 at the shipped 256 MiB tier.

1. The C++ toolchain does not fit. Eight runs returned a bare SE with no case results and zero CPU time, all of them C++: the container never produced a compiled binary. Measuring the compiler directly, `g++ -O2` peaks at 189–211 MB of resident memory on these sources (P1 189.4 MB, P2 190.9 MB, P3 198.2 MB, P4 210.8 MB). Compilation runs inside the submission container, so a 128 MiB tier cannot build them at all. This is a property of the tier, not of the scheduler: no strategy can rescue a submission that never compiled. The knapsack source compiles in 60.6 MB and so does reach execution, which is why C++ shows both SE and MLE rather than one or the other.

2. The watermark window is too narrow. Every knapsack cell failed in every language under every strategy, including

Reactive and Hybrid, which did promote. Promotion completed at 253–794 ms, but the program crosses 128 MiB before it can land. The window between the watermark (89.6 MiB) and the hard limit is only 38.4 MiB, and these submissions allocate at a rate that consumes it well inside the promotion path’s latency. At the 256 MiB tier the same window is 76.8 MiB and the mechanism holds for C, C++ and Python.

3. The JVM heap allowance contradicts the container cap. Java was the only language to fail at the shipped tier as well: three MLE verdicts at 256 MiB, in Predictive, Reactive and Hybrid alike, with peaks of 255–256 MB. The runtime is started with `-Xmx512m`, a 512 MB heap allowance, inside a 256 MiB container. The JVM grows its heap toward that allowance and crosses the container ceiling. The two limits are mutually inconsistent and should be reconciled by sizing `-Xmx` as a fraction of the tier, or by raising the tier for JVM workloads.

Taken together these set **256 MiB as the practical minimum Tier 1 limit**, but for reasons different from those we initially expected. The limit is not set by the classifier, and only partly by the promotion mechanism: it is set by the largest of three demands the tier must simultaneously satisfy, namely the toolchain used to build the submission, the watermark window needed to promote it in time, and the runtime’s own overhead above the program’s working set.

A note on how the tier was enforced. Our first 128 MiB run reported that C, C++ and Python all passed, and we initially read this as evidence that 128 MiB was safe for native and managed languages alike. It was not. The judge set `--memory` on the container but not `--memory-swap`, and Docker defaults the swap limit to the same value as `--memory`, so the effective ceiling was roughly twice the nominal tier. A probe that allocates and touches a chosen footprint survived 235 MB and was killed at 245 MB under a nominal 128 MiB tier. Passing swap equal to memory removes the allowance, after which the same probe survives 120 MB and is killed at 130 MB. The figures in this subsection are from runs with the tier genuinely enforced. The same default also inflated every peak reported for a capped run, since `memory.current` cannot exceed `memory.max` and the peak is sampled from `memory.current`: a capped run reports its own limit as its peak, so true demand is only observable in uncapped runs.

B. Future Research Scopes

Based on these boundary observations, we outline five key future scopes:

- 1) **Toolchain-Aware Tier Sizing:** A 128 MiB tier cannot compile C++ at all, since `g++` alone peaks at 189–211 MB, so tier selection must account for the build step and not only for the runtime. Measuring per-language build cost and provisioning the tier as `max(build, runtime)` would let lighter languages run in a smaller tier without stranding compiled ones.
- 2) **Watermark Tuning Against Allocation Rate:** The 38.4 MiB window at a 128 MiB tier is too narrow

to outrun a submission needing twice the tier, whereas 76.8 MiB holds at 256 MiB. What matters is not the fraction but the absolute window relative to how fast a submission claims memory, which argues for sizing the watermark from a measured allocation rate rather than a fixed percentage.

- 3) **Reconciling `-Xmx` with the Tier:** The runtime is configured with a 512 MB heap allowance inside a 256 MiB container, which cannot be honoured. Sizing `-Xmx` below the tier, and pairing it with an allocator that grows heap in smaller increments such as `-XX:+UseSerialGC`, would remove the contradiction we measured.
- 4) **Deliberate, Bounded Swap:** We disabled swap because Docker’s default was silently doubling the tier and invalidating our measurements. A small, explicitly sized allowance (e.g. `--memory-swap 25% above --memory`) could absorb the allocation spikes that occur during promotion, provided the tier’s real ceiling is stated rather than left to a default.
- 5) **Hardware-Level MicroVM Isolation:** Integrating Firecracker or gVisor to provide kernel-level security hardening for the Heavy execution tier.

VII. CONCLUSION

Fixed, worst-case resource limits reduce how many submissions an online judge can run concurrently, and that effect is most visible on fixed physical hardware where more machines cannot simply be added. In our simulations, seeded with real measured profiles, starting every submission in a bounded 256 MiB tier and promoting only on measured kernel pressure reclaimed 47.22% of reserved memory and 22.09% of reserved CPU core-hours relative to a static 2048 MiB baseline. The simulated 500-submission burst showed average queue wait falling from 12,386.2 ms to 0.0 ms, with a correspondingly shorter clear time.

Three findings temper the result, and we consider all three load-bearing rather than incidental.

First, prediction did not pay for itself on this corpus. Routing by the classifier alone reached 40.23%, and prediction combined with the watermark reached only 22.45%, because a container the classifier marks heavy is left uncapped and therefore carries no reserved capacity to count as saving. Since the two strategies share one classifier for the initial tier and promotion can only raise a reservation, Hybrid can never reserve less than Predictive: it pays for the classifier’s false alarms and still needs the watermark. Reactive promotion, which commits little and escalates on evidence, was the most efficient strategy measured, at 47.22%. That the mechanism outperforms the predictor is the paper’s more defensible claim, and it points to classifier reuse rather than classifier reliance as the design direction.

Second, the classifier is weak outside its training distribution. It routes 33.3% of submissions correctly on our six-problem corpus, below the 77.8% a constant *light* prediction would score, despite reaching 94.45% held-out accuracy on

CodeNet. A system whose resource safety depended on that classifier would be fragile; the watermark does not depend on it, which is what makes a bounded-tier start safe to deploy.

Third, our attempt to locate the floor by squeezing Tier 1 to 128 MiB failed for three independent reasons, and we report that failure rather than a tidier positive result, because it is what establishes the floor. The C++ toolchain alone needs 189–211 MB and cannot build a submission inside the tier; the 38.4 MiB watermark window is too narrow to promote a submission that outgrows the tier, so every strategy lost the memory-heavy problem; and the JVM is configured with a heap allowance twice the size of its container, an inconsistency that surfaces as MLE at 256 MiB as well. The tier must therefore simultaneously accommodate the build step, the watermark window needed for timely promotion, and the runtime’s overhead above the program’s working set. It is the largest of those three, not the classifier, that sets 256 MiB as the practical minimum.

Taken together, these results indicate that adaptive resource allocation is a practical direction for online code judge infrastructure, with the watermark rather than the classifier as its load-bearing component. The cloud cost figures remain a projection, and the approach still needs testing against a wider range of real, larger-scale workloads.

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